4CS015 – Portfolio Part -4

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You will need to complete the workshop and then submit this Word file, complete with your answers, via Canvas, by the end of Semester (week 12).

For this assignment (workshop) you will show you how to navigate and create file systems from command prompt in both Windows (os) and Linux (os), also how to create a batch file to be executed in the Windows (os).

NOTE: Please make sure you have written your command as well as the screenshot of the commands. Failure to do so will result in marks deduction.

Table of Contents

4CS015 – Portfolio Part -4 ..... **1**

Task 1 (Windows Command Line) ***2***

Task 2 ..... ***3***

Task 3 (Linux terminal) ***2***

Task 4 ..... ***3***

Task 5 ..... ***5***

Task 1 (Windows Command Line)

1. Write a command to Create a directory "**test**"

   Ans: mkdir test

2. Write the command to list all the directories/files inside the directory **test**

   Ans: dir test

3. Create a file "**hello.txt**" containing the word "Hello", inside the test directory.

   Ans: echo Hello > test\hello.txt

4. Display the content of **hello.txt** in the command line.

   Ans: type test\hello.txt

5. Make directory **test1** in the Desktop and copy **hello.txt** from **test** to **test1**

   Ans: **mkdir %USERPROFILE%\Desktop\test1**

   copy test\hello.txt %USERPROFILE%\Desktop\test1

6. Now delete the file **hello.txt** in the current directory

   Ans: del hello.txt

7. Write a command to display your current directory.

   Ans: cd

8. Now rename the file to **hello2.txt**

   Ans: rename hello.txt hello2.txt

9. Now navigate outside the directory test.

   Ans: cd ..

10. Write the command to print the tree structure. It should look something like this:

```
Folder PATH listing for volume Offline
Volume serial number is 0000-0000
U:\TEST
└───test2
        hello2.txt
```

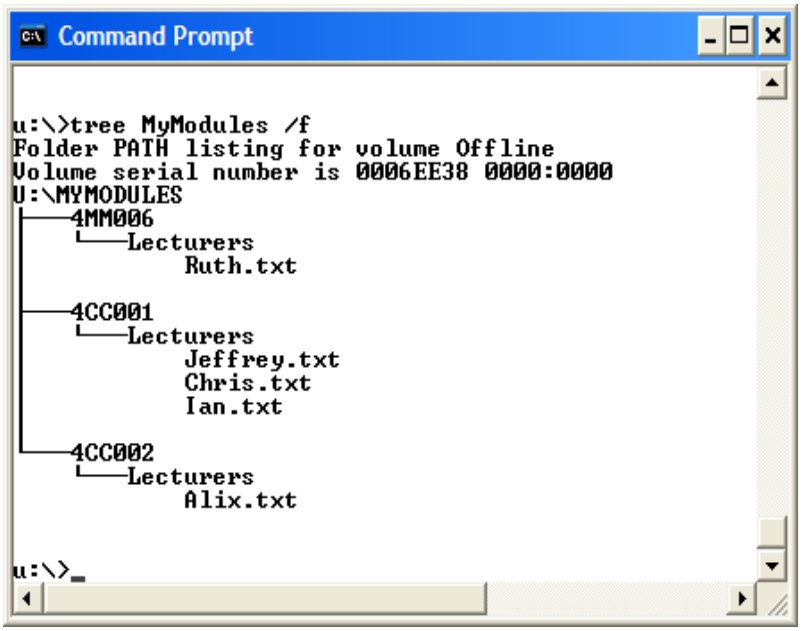
   Ans: tree /f

Task 2

Create the following directory structure and paste it after this picture. (15 marks):

Note: MYMODULES must be inside <your_name> directory.

Make a directory with your name on the desktop and navigate to the directory before you display the following structure.



cd %USERPROFILE%\Desktop
mkdir YourName
cd YourName
mkdir MYMODULES
cd MYMODULES
mkdir 4MM006
mkdir 4CC001
mkdir 4CC002
mkdir 4MM006\Lecturers
mkdir 4CC001\Lecturers
mkdir 4CC002\Lecturers
type nul > 4MM006\Lecturers\Ruth.txt
type nul > 4CC001\Lecturers\Jeffrey.txt
type nul > 4CC001\Lecturers\Chris.txt
type nul > 4CC001\Lecturers\Ian.txt
type nul > 4CC002\Lecturers\Alix.txt
tree MYMODULES /f
(dir /s /b)

```
C:\users\crossover\Desktop\sagar\MYMODULES>dir /s /b
C:\users\crossover\Desktop\sagar\MYMODULES\4CC001
C:\users\crossover\Desktop\sagar\MYMODULES\4cc002
C:\users\crossover\Desktop\sagar\MYMODULES\4MM006
C:\users\crossover\Desktop\sagar\MYMODULES\4CC001\Lectures
C:\users\crossover\Desktop\sagar\MYMODULES\4CC001\Lectures\Chris.txt
C:\users\crossover\Desktop\sagar\MYMODULES\4CC001\Lectures\Jeffrey.txt
C:\users\crossover\Desktop\sagar\MYMODULES\4CC001\Lectures\lan.txt
C:\users\crossover\Desktop\sagar\MYMODULES\4cc002\Lectures
C:\users\crossover\Desktop\sagar\MYMODULES\4cc002\Lectures\Alix.txt
C:\users\crossover\Desktop\sagar\MYMODULES\4MM006\Lectures
C:\users\crossover\Desktop\sagar\MYMODULES\4MM006\Lectures\Ruth.txt
```

Task 3 (Linux terminal) (10 marks)

Now, Boot into Linux operating system in virtualbox or Windows-SubSystem for LInux(On Windows)

1. Write a command to Create a directory "**test**"

   Ans: mkdir test

2. Write the command to list all the directories/files inside the directory **test**

   Ans: ls test

3. Create a file "**hello.txt**" containing the word "Hello", inside the test directory.

   Ans: echo “Hello” > test/hello.txt

4. Display the content of **hello.txt** in the command line.

   Ans: cat test/hello.txt

5. Make directory **test1** in the Desktop and copy **hello.txt** from **test** to **test1**

   Ans: mkdir ~/Desktop/test1

   cp test/hello.txt ~/Desktop/test1/

6. Now delete the file **hello.txt** in the current directory

   Ans: rm test/hello.txt

7. Write a command to display your current directory.

   Ans: pwd

8. Now rename the file to **hello2.txt**

   Ans: mv hello.txt hello2.txt

9. Now navigate outside the directory test.

   Ans: cd . .

10. Write the command to print the tree structure.

    Ans: tree

Task 4

Create the following directory structure (identical to the one that you created in Windows Command Prompt):

Note: MYMODULES must be inside <your_name> directory.

```
ting@ubuntu-desktop: ~
File  Edit  View  Terminal  Help
ting@ubuntu-desktop:~$ ls -R MyModules
MyModules:
4cc001  4cc002  4mm006

MyModules/4cc001:
Lecturers

MyModules/4cc001/Lecturers:
Chris.txt  Ian.txt  Jeffrey.txt

MyModules/4cc002:
Lecturers

MyModules/4cc002/Lecturers:
Alix.txt

MyModules/4mm006:
Lecturers

MyModules/4mm006/Lecturers:
Ruth.txt
ting@ubuntu-desktop:~$
```

and insert it below (15 marks):

[Insert your picture here]

```
[sagarmishra841@Sagars-MacBook-Air-2 ~ % cd ~/desktop
[sagarmishra841@Sagars-MacBook-Air-2 desktop % mkdir sagar
[sagarmishra841@Sagars-MacBook-Air-2 desktop % cd sagar
[sagarmishra841@Sagars-MacBook-Air-2 sagar % mkdir MyModules
[sagarmishra841@Sagars-MacBook-Air-2 sagar % cd MyModules
[sagarmishra841@Sagars-MacBook-Air-2 MyModules % mkdir 4cc001 4cc002 4mm006
[sagarmishra841@Sagars-MacBook-Air-2 MyModules % mkdir 4cc001/Lecturers
[sagarmishra841@Sagars-MacBook-Air-2 MyModules % mkdir 4cc002/Lecturers
[sagarmishra841@Sagars-MacBook-Air-2 MyModules % mkdir 4mm006/Lecturers
[sagarmishra841@Sagars-MacBook-Air-2 MyModules % touch 4cc001/Lecturers/Chris.txt
[sagarmishra841@Sagars-MacBook-Air-2 MyModules % touch 4cc001/Lecturers/Ian.txt
[sagarmishra841@Sagars-MacBook-Air-2 MyModules % touch 4cc001/Lecturers/Jeffrey.txt
[sagarmishra841@Sagars-MacBook-Air-2 MyModules % touch 4cc002/Lecturers/Alix.txt
[sagarmishra841@Sagars-MacBook-Air-2 MyModules % touch 4mm006/Lecturers/Ruth.txt
[sagarmishra841@Sagars-MacBook-Air-2 MyModules % ls -R MyModules
ls: MyModules: No such file or directory
[sagarmishra841@Sagars-MacBook-Air-2 MyModules % cd
[sagarmishra841@Sagars-MacBook-Air-2 ~ % cd ~/sektop
cd: no such file or directory: /Users/sagarmishra841/sektop
[sagarmishra841@Sagars-MacBook-Air-2 ~ % cd ~/desktop
[sagarmishra841@Sagars-MacBook-Air-2 desktop % ls -R MyModules
ls: MyModules: No such file or directory
[sagarmishra841@Sagars-MacBook-Air-2 desktop % cd sagar
[sagarmishra841@Sagars-MacBook-Air-2 sagar % ls -R MyModules
4cc001  4cc002  4mm006

MyModules/4cc001:
Lecturers

MyModules/4cc001/Lecturers:
Chris.txt       Ian.txt         Jeffrey.txt

MyModules/4cc002:
Lecturers

MyModules/4cc002/Lecturers:
Alix.txt

MyModules/4mm006:
Lecturers

MyModules/4mm006/Lecturers:
Ruth.txt
sagarmishra841@Sagars-MacBook-Air-2 sagar %
```

Task 5

Batch file material (50 marks in total)

Create a batch file (windows) **YourNumber.bat** from notepad (or your preferred text editor).

The batch file should create a directory structure similar to the one create for the first section of this assignment but the directory entries should be provided from an input request from the batch file itself.

You should be able to create a root directory and directory structure and populate it if required. The batch file program will ask for names (**input directory names and file names**) to be used and have some form of **error checking** (empty – incorrect character, length, etc.) You can add other features such as being able to edit the new structure from within the batch file program (**find and change a directory name**).

Run batch file from windows prompt: YourNumber.bat <enter>

[Insert your picture(s) here] – Directory structure created / menu of batch file (inputs and output when batch file is executed) (10 marks)

```
Microsoft Windows 10.0.22000

Z:\Users\sagarmishra841\Library\Containers\com.isaacmarovitz.Whisky\Bottles\B67C
FA79-5F77-4AB4-9D19-DACE354784B5\drive_c\windows\system32>cd ..

Z:\Users\sagarmishra841\Library\Containers\com.isaacmarovitz.Whisky\Bottles\B67C
FA79-5F77-4AB4-9D19-DACE354784B5\drive_c\windows>cd ..

Z:\Users\sagarmishra841\Library\Containers\com.isaacmarovitz.Whisky\Bottles\B67C
FA79-5F77-4AB4-9D19-DACE354784B5\drive_c>cd batch

Z:\Users\sagarmishra841\Library\Containers\com.isaacmarovitz.Whisky\Bottles\B67C
FA79-5F77-4AB4-9D19-DACE354784B5\drive_c\batch>2606063.bat
```

```
==========================================
       DIRECTORY STRUCTURE CREATOR
==========================================
1. Create new directory structure
2. Rename an existing directory
3. Exit
==========================================
Enter your choice (1-3): _
```

```
Enter root directory name: sagar
Enter first subdirectory name: 2606063
Enter second subdirectory name:
Enter file name to create (with extension):

Directory structure created successfully!
Press any key to continue...
```

```
Enter existing directory name: sagar
Enter new directory name: sagar mishra
Directory renamed successfully!
Press any key to continue...
```

[Insert your text here] – Batch file listing (script)(40 marks)

```
@echo off

title Dynamic Directory Structure Creator

color 0A

:MENU

cls

echo ==========================================

echo       DIRECTORY STRUCTURE CREATOR

echo ==========================================

echo 1. Create new directory structure

echo 2. Rename an existing directory

echo 3. Exit

echo ==========================================

set /p choice=Enter your choice (1-3):

if "%choice%"=="1" goto CREATE

if "%choice%"=="2" goto RENAME

if "%choice%"=="3" exit

echo Invalid choice!

pause

goto MENU

:CREATE

cls

set /p ROOT=Enter root directory name:

:: Error checking for empty input

if "%ROOT%"=="" (

   echo Root directory name cannot be empty!

   pause

   goto CREATE

)

mkdir "%ROOT%" 2>nul

if errorlevel 1 (

   echo Directory already exists or invalid name!

   pause

   goto MENU

)

cd "%ROOT%"

set /p D1=Enter first subdirectory name:

if "%D1%"=="" goto SKIP1

mkdir "%D1%"

:SKIP1

set /p D2=Enter second subdirectory name:

if "%D2%"=="" goto SKIP2

mkdir "%D2%"

:SKIP2

set /p FILE1=Enter file name to create (with extension):

if "%FILE1%"=="" goto DONE

echo This is a sample file > "%FILE1%"

:DONE

echo.
```

```
echo Directory structure created successfully!
cd ..
pause
goto MENU
:RENAME
cls
set /p OLD=Enter existing directory name:
if not exist "%OLD%" (
   echo Directory does not exist!
   pause
   goto MENU
)
set /p NEW=Enter new directory name:
if "%NEW%"== "" (
   echo New name cannot be empty!
   pause
   goto MENU
)

ren "%OLD%" "%NEW%"
echo Directory renamed successfully!
pause
goto MENU
```